EE4047 – High Voltage Techniques I

New Topic:

Approximated Electric Fields

Week 11

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OUTLINE

1 Plane-Sphere Electrode System

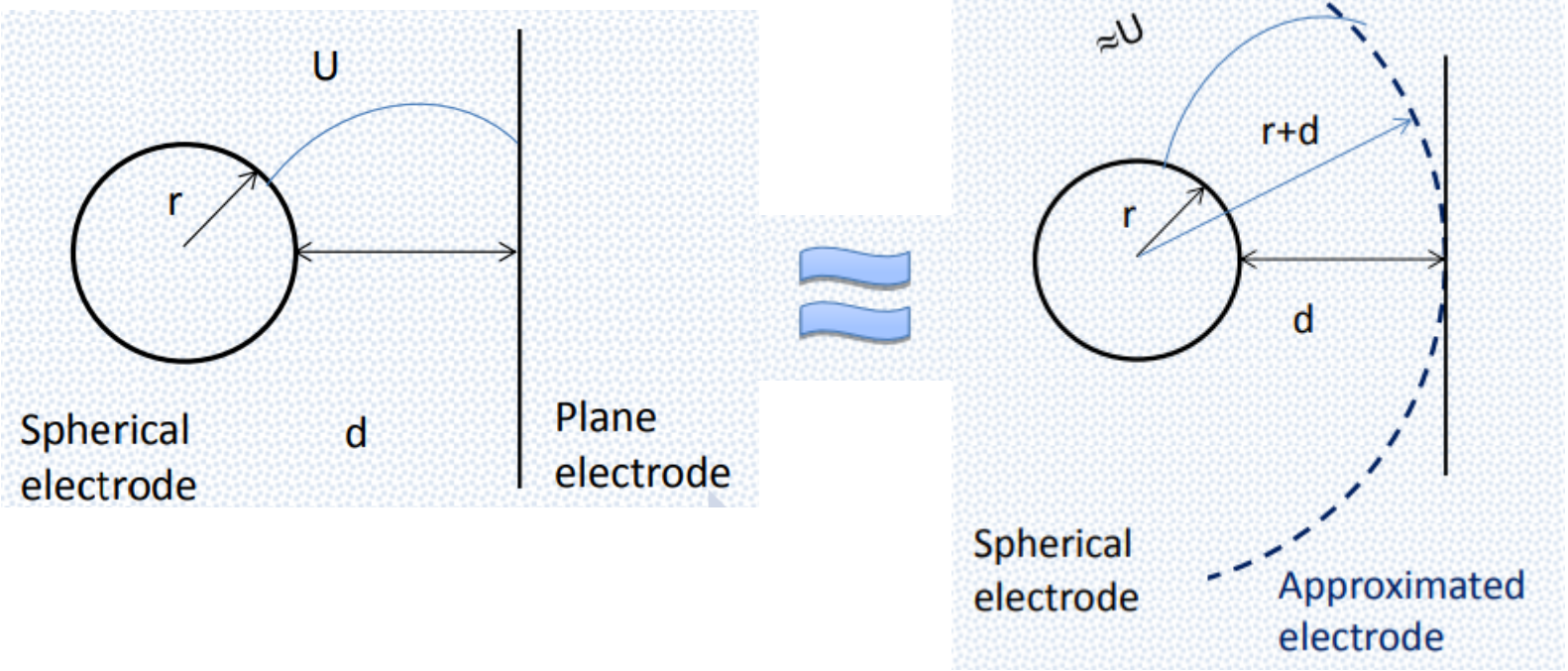
2 Sphere-Sphere Electrode System

3 Plane-Cylinder Electrode System

4 Cylinder-Cylinder Electrode System

1 PLANE-SPHERE ELECTRODE SYSTEM

Plane-sphere electrode system



PLANE-SPHERE ELECTRODE SYSTEM

Plane-sphere electrode system

$$E_{max} = \frac{U}{\frac{r_1}{r_2}(r_2 - r_1)}$$

$r_1 = r$, $r_2 = r + d$, $U = U$

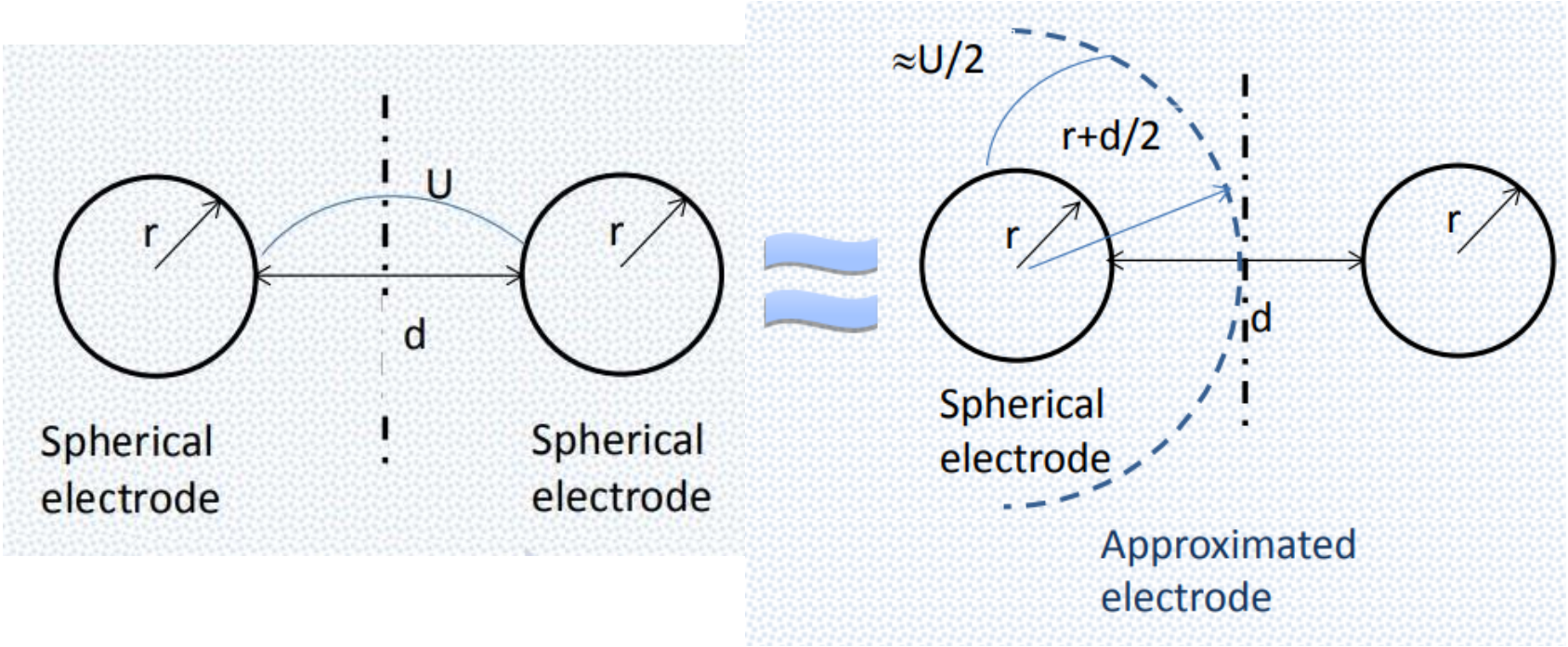
$$E_{max} = \frac{U}{\frac{r}{r + d}(r + d - r)}$$

$$E_{max} = \boxed{k}\frac{U}{d}.\frac{r + d}{r}$$

k ≅ 0.85 - 0.95

SPHERE-SPHERE ELECTRODE SYSTEM

Sphere-sphere electrode system



SPHERE-SPHERE ELECTRODE SYSTEM

Sphere-sphere electrode system

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$r_1 = r,\ r_2 = r + d/2,\ U = U/2$

$$E_{max} = \frac{U/2}{\frac{r}{r + d/2}(r + d/2 - r)}$$

$$E_{max} = \boxed{k}\frac{U}{d}.\frac{r + d/2}{r}$$

$k \cong 0.85 - 0.95$

2 SPHERE-SPHERE ELECTRODE SYSTEM

Example:

In a sphere-sphere (identical spheres) electrode system, the breakdown voltage is 13 kV. The correction factor in approximated field calculation is k=0.95 and breakdown strength of air is supposed to be E_b =30 $^{kV/cm}$. Determine radii of the spheres (r) and electrode separation (d) when r = 10 d.

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$$E_b = K \frac{U_b/2}{\frac{r}{r+d/2}(r+d/2-r)} = K \frac{U_b}{2} \frac{2r+d}{dr} \quad \& \quad r=10d$$

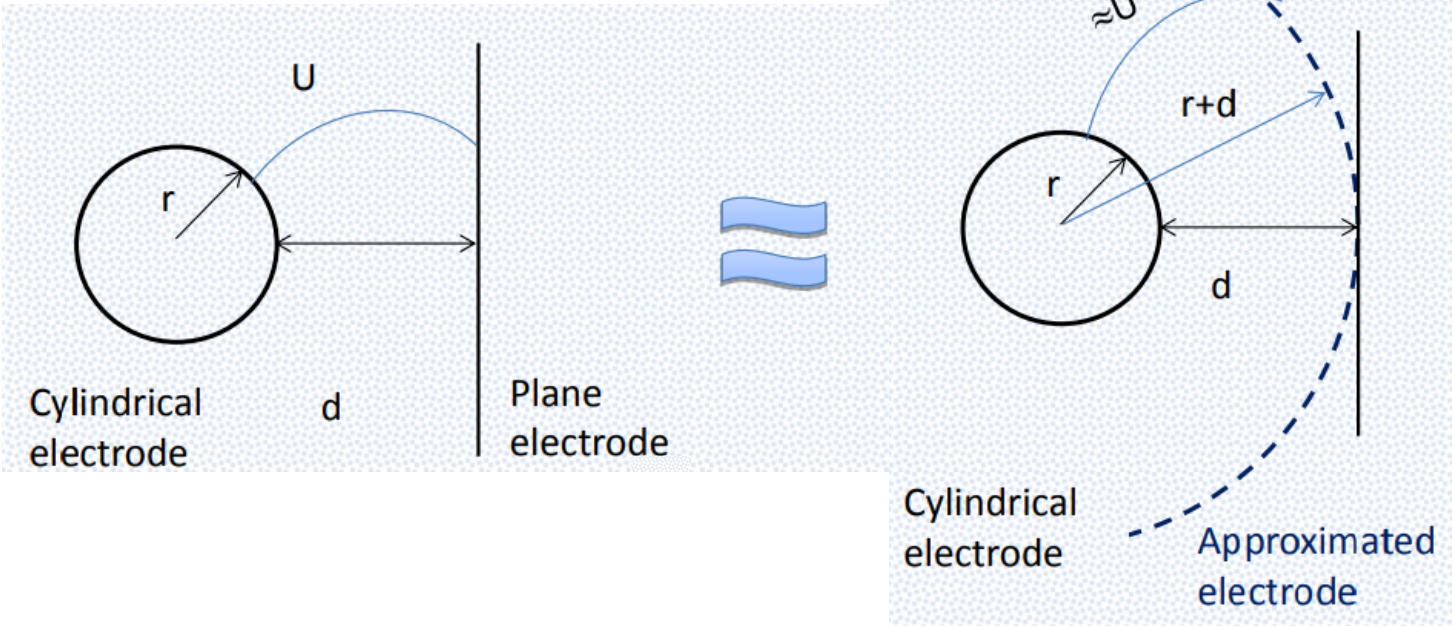


$$30 \frac{kV}{cm} = 0.95 \frac{13}{2} \frac{2(10d)+d}{d(10d)} = 6.175 \frac{21d}{10d^2} = \frac{12.9675}{d} \rightarrow d = 0.43225\,cm$$

$$r = 10d = 4.3225$$

PLANE-CYLINDER ELECTRODE SYSTEM

Plane-cylinder electrode system



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Plane-cylinder electrode system

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$r_1 = r,\ r_2 = r + d,\ U = U$

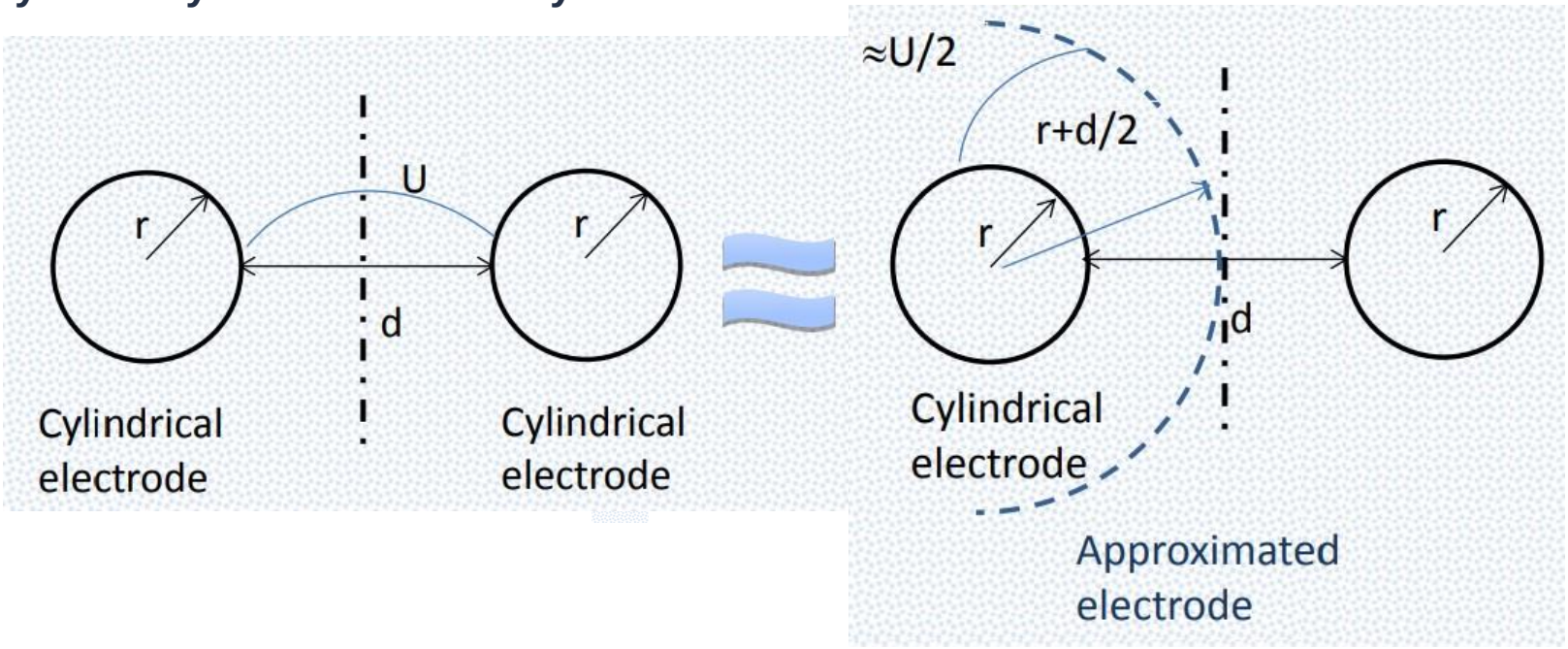
$$E_{max} = \frac{U}{r \ln \frac{r + d}{r}}$$

$$E_{max} = k \frac{U}{r \ln \frac{r + d}{r}}$$

$k \cong 0.85 - 0.95$

CYLINDER-CYLINDER ELECTRODE SYSTEM

Cylinder-cylinder electrode system



CYLINDER-CYLINDER ELECTRODE SYSTEM

Cylinder-cylinder electrode system

$$E_{max} = \frac{U}{r_1 \ln \frac{r_2}{r_1}}$$

$r_1 = r,\ r_2 = r + d/2,\ U = U/2$

$$E_{max} = \frac{U/2}{r \ln \frac{r + d/2}{r}}$$

$$E_{max} = k \frac{U}{2r \ln \frac{r + d/2}{r}}$$

$k \cong 0.85 - 0.95$

CYLINDER-CYLINDER ELECTRODE SYSTEM

Example:

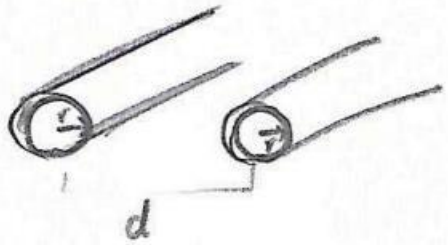
In a three phase 380 ^{kV} transmission line, diameter of the conductors are 32 ^{mm}. The breakdown strength of air is 30 $^{kV/cm}$ and the correction factor is k=0.85. Using approximated maximum field calculation, determine the minimum distance between two phases.

CYLINDER-CYLINDER ELECTRODE SYSTEM

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$r = \frac{32}{2} = 16^{mm} = 1.6^{cm}$



$E_{MAX} = k \frac{U/2}{r \ln\left(\frac{r+d/2}{r}\right)} \rightarrow 30 \frac{kV}{cm} = 0.85 \frac{380/2}{1.6 \ln\left(\frac{1.6+\frac{d}{2}}{1.6}\right)}$

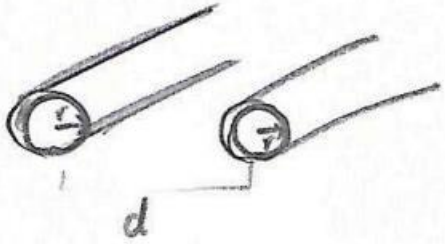
$\rightarrow \ln\left(\frac{1.6+d/2}{1.6}\right) = \frac{0.85 \times 190}{1.6 \times 30} = 3.365$

CYLINDER-CYLINDER ELECTRODE SYSTEM

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$$\frac{1.6 + d/2}{1.6} = e^{3.365} = 28.92 \rightarrow 1.6 + \frac{d}{2} = 46.272$$

$$d = 89.34^{cm}$$

END OF THE LESSON

Any questions?